

Chapter 19: Compatibility of Artificial Intelligence in Field Research: A Sustainable Development Perspective

A Conceptual and Scoping Review

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Abstract

Field research — the systematic collection of primary data in natural, agricultural, ecological, or community settings — is increasingly being reshaped by artificial intelligence (AI). This paper presents a conceptual and scoping review examining the compatibility of AI technologies with field-research practice through the lens of sustainable development. Drawing on recent bibliometric, systematic, and empirical studies indexed in Scopus, Web of Science, and related databases, the review synthesises evidence on how machine learning, computer vision, natural language processing, and AI-enabled remote sensing intersect with field-based data collection across environmental, agricultural, social, and public-health domains. The analysis is organised around four dimensions of compatibility — technical, economic, ethical-social, and institutional-policy — and maps AI-enabled field-research applications onto the United Nations Sustainable Development Goals (SDGs). The review finds that AI substantially improves the scale, speed, and precision of field data acquisition and analysis, but that its compatibility with sustainable development is conditional rather than automatic: infrastructural gaps, algorithmic bias, data-privacy risks, and the ecological footprint of AI systems themselves can offset its benefits, particularly in low-resource settings. The paper concludes with a set of recommendations for researchers, institutions, and policymakers seeking to align AI-enabled field research with the principles of sustainability, equity, and scientific integrity, and identifies priority areas for future empirical investigation.

Keywords: artificial intelligence; field research; sustainable development; Sustainable Development Goals; remote sensing; environmental monitoring; digital divide; research ethics

1. Introduction

Field research constitutes the empirical backbone of the natural, environmental, agricultural, and social sciences, providing the primary data upon which theory, policy, and practice are built. Traditionally reliant on manual observation, physical sampling, and researcher-administered instruments, field research has historically been constrained by cost, personnel availability, accessibility of remote or hazardous sites, and the sheer volume of data that human observers can process. Over the past decade, advances in artificial intelligence (AI) — including machine learning (ML), deep learning (DL), computer vision, and natural language processing (NLP) — have begun to reconfigure how field data are collected, processed, and interpreted.

At the same time, the 2030 Agenda for Sustainable Development has established a global normative framework of seventeen interlinked Sustainable Development Goals (SDGs) that increasingly shape the priorities of scientific research, including field-based investigation. A growing body of literature argues that AI can accelerate progress toward the SDGs by enabling systems-level, data-driven insight across environmental, social, and economic domains; one influential expert-elicitation study estimates that AI can support 134 of the 169 SDG targets, while potentially inhibiting 59 of them, underscoring that its contribution is neither uniform nor guaranteed.

This raises a central question that motivates the present paper: to what extent — and under what conditions — is AI compatible with the practice of field research when judged against the goals and principles of sustainable development? Compatibility, in this sense, is understood not merely as technical feasibility, but as the degree to which AI-enabled field-research methods can be adopted, sustained, and scaled without generating disproportionate ecological, economic, or social costs. This paper addresses the question through a conceptual and scoping review of recent literature, organised around technical, economic, ethical-social, and institutional dimensions of compatibility, and mapped explicitly onto the SDG framework.

2. Conceptual Background and Literature Review

2.1 Artificial Intelligence and the Sustainable Development Agenda

The relationship between AI and sustainable development has attracted substantial scholarly attention. Vinuesa et al. conducted a consensus-based expert elicitation and found that AI can act as an enabler across nearly all seventeen SDGs, while simultaneously cautioning that unregulated deployment can inhibit a significant number of targets, particularly those concerned with equity, decent work, and environmental integrity. Building on this foundation, subsequent bibliometric analyses of AI-and-sustainability literature — spanning thousands of Scopus- and Web of Science-indexed documents — report exponential growth in publication output since roughly 2015, with machine learning and deep learning identified as the dominant methodological drivers of SDG-relevant applications. A recent large-scale review of 792 articles on AI applications in SDG-related research similarly finds that the field spans a wide disciplinary spectrum, from the natural sciences to the humanities, but identifies a persistent gap: relatively few studies combine advanced AI methods with deep substantive sustainability expertise, suggesting that technical capability has outpaced integrative, domain-grounded application.

Complementary work has examined AI's role through the lens of the general-purpose technology concept, arguing that AI can catalyse innovation across sectors while also generating governance and ethical challenges that must be actively managed rather than assumed away. Ethical-design frameworks proposed for AI-enabled research and development emphasise dual accountability for both process-related and outcome-related sustainability, invoking principles of beneficence and non-maleficence toward all research stakeholders.

2.2 Field Research Methodologies and the Turn Toward AI-Enabled Data Collection

Field research spans a heterogeneous set of methodologies, including ecological and biodiversity surveys, environmental and pollution monitoring, precision and smallholder agriculture, remote sensing and Earth observation, and social or behavioural fieldwork involving interviews, surveys, and ethnographic observation. Across each of these domains, recent reviews document a shift from manual, labour-intensive protocols toward AI-augmented pipelines. In environmental monitoring, reviews spanning several thousand publications report that AI and machine learning are increasingly integrated with the Internet of Things (IoT) and remote sensing to enable real-time data collection, automated anomaly detection, and predictive modelling across air quality, water quality, biodiversity, and disaster-management applications. In tropical forest monitoring specifically, AI and machine learning have been integrated into remote-sensing pipelines precisely because conventional field inventories are constrained

by limited or uneven ground data, persistent cloud cover, and limited institutional capacity — conditions that make AI-based remote inference an attractive complement to, rather than a wholesale replacement for, on-the-ground fieldwork.

A field trial conducted with the Wisconsin Department of Natural Resources offers a concrete illustration: a computer-vision pipeline applied to remote-sensing imagery reduced the volume of manually reviewed images by 99.8 percent while substantially increasing the likelihood that flagged images contained genuine environmental violations, demonstrating how AI can convert an otherwise intractable field-monitoring burden into a manageable one. Comparable dynamics are reported in reviews of AI in remote sensing and satellite image processing more broadly, where deep-learning techniques are applied to land-cover classification, object detection, and climate-change monitoring at scales unattainable through manual field survey alone.

3. Methodology

This paper adopts a conceptual and scoping review methodology, appropriate for synthesising a heterogeneous and rapidly evolving evidence base rather than testing a narrowly defined hypothesis. The review draws on peer-reviewed articles, systematic and bibliometric reviews, and grey-literature reports identified through targeted searches of Scopus-indexed and related academic sources, using combinations of the terms 'artificial intelligence', 'field research', 'field study', 'remote sensing', 'environmental monitoring', 'sustainable development', and 'Sustainable Development Goals'. Sources were prioritised for inclusion where they (a) explicitly examined AI applications within field-based or field-adjacent data collection contexts, (b) addressed compatibility, adoption barriers, or sustainability implications of AI deployment, or (c) provided conceptual or empirical linkage between AI and the SDG framework. The synthesis is organised thematically around four compatibility dimensions — technical, economic, ethical-social, and institutional-policy — which were derived inductively from recurring themes across the reviewed literature, and is complemented by a mapping exercise linking field-research domains to relevant SDGs.

4. AI Applications Enhancing Compatibility with Field Research

Table 1 summarises representative AI techniques applied across major field-research domains and their principal linkages to the SDG framework, synthesised from the reviewed literature.

Table 1. AI-Enabled Field Research Domains and Corresponding SDG Linkages

Field Research Domain	Representative AI Techniques	Primary SDG Linkages
Ecological and biodiversity surveys	Computer vision for camera-trap species identification; bioacoustic classifiers; citizen-science data fusion	SDG 14, 15
Environmental and pollution monitoring	Sensor-network anomaly detection; deep learning on satellite/UAV imagery for land-cover and emissions tracking	SDG 6, 11, 13

Field Research Domain	Representative AI Techniques	Primary SDG Linkages
Precision and smallholder agriculture	Predictive analytics for yield and irrigation; drone-based crop-health imaging; pest and disease recognition models	SDG 2, 12, 13
Remote sensing and Earth observation	Convolutional neural networks for change detection; forest and deforestation monitoring; disaster-response mapping	SDG 13, 15, 11
Social and behavioural fieldwork	Natural-language processing for interview transcription and thematic coding; adaptive survey instruments; mobile sensing	SDG 1, 4, 10
Public-health and epidemiological field studies	Predictive outbreak modelling; AI-assisted diagnostics in mobile clinics; data triage for community health workers	SDG 3

4.1 Environmental and Ecological Field Research

AI-enabled camera traps, bioacoustic monitoring, and computer-vision species classifiers are increasingly used to automate biodiversity surveys that would otherwise require extensive manual observation. Reviews of environmental remote sensing report that machine learning and deep learning are now applied to water, agricultural, urban, and wetland monitoring, with foundation models beginning to extend these capabilities further. Such tools do not eliminate the need for field presence but substantially reduce the volume of raw observational data that must be manually processed, freeing researcher time for validation, interpretation, and higher-order analysis.

4.2 Agricultural Field Research

In agricultural field research, AI supports precision irrigation, yield forecasting, and pest and disease detection through drone- and sensor-based imagery, with direct relevance to SDG 2 (zero hunger) and SDG 12 (responsible consumption and production). These applications are particularly significant for smallholder and resource-constrained farming systems, where AI-enabled early-warning tools can substitute for the intensive extension services that many agricultural systems lack.

4.3 Remote Sensing and Earth Observation

The integration of AI with remote sensing has been described as having revolutionised Earth observation by enabling automated, efficient, and precise analysis of vast and complex datasets drawn from satellite imagery, aerial photography, and ground-based sensors. Comprehensive reviews spanning literature indexed in Scopus, Web of Science, and IEEE Xplore report applications across land-cover mapping,

change detection, and disaster-response mapping, with particular value in contexts where field access is limited by terrain, cost, or political instability.

4.4 Social, Behavioural, and Public-Health Field Research

Beyond the environmental and agricultural sciences, AI-enabled natural language processing supports transcription, coding, and thematic analysis of qualitative field data, while adaptive digital survey instruments and mobile sensing extend the reach of social-science fieldwork into hard-to-access populations. In public-health field research, predictive outbreak modelling and AI-assisted diagnostic triage have been used to extend the effective reach of community health workers operating in resource-constrained settings, with direct relevance to SDG 3 (good health and well-being).

5. Dimensions of Compatibility

5.1 Technical Compatibility

Technical compatibility concerns the extent to which AI systems can be reliably deployed within the physical and data conditions of field settings — including intermittent connectivity, variable environmental conditions, and heterogeneous, often noisy, sensor data. Reviews of AI in environmental remote sensing note persistent challenges of model interpretability, multimodal data integration, and the need for real-time deployment at global scale, indicating that technical maturity varies considerably across application domains.

5.2 Economic Compatibility

Economic compatibility concerns cost, scalability, and return on investment relative to conventional field methods. Evidence from applied deployments — such as the substantial reduction in manual image-review workload achieved in AI-assisted environmental compliance monitoring — suggests that, once developed, AI pipelines can dramatically lower the marginal cost of large-scale field monitoring. However, the upfront costs of sensor infrastructure, computing capacity, and specialised expertise remain substantial barriers, particularly for research institutions in low- and middle-income countries.

5.3 Ethical and Social Compatibility

Ethical compatibility concerns data privacy, informed consent, algorithmic bias, and the distribution of benefits and risks among research participants and host communities. Ethical frameworks for AI-enabled research emphasise dual accountability to both the research process and its outcomes, requiring that AI-enabled field studies uphold beneficence toward, and avoid harm to, the individuals, communities, and ecosystems they study. Survey-based evidence on AI adoption in sustainable-development research corroborates this concern: a majority of surveyed practitioners identify data privacy and security, alongside the absence of clear AI governance strategy, as leading barriers to responsible adoption.

5.4 Institutional and Policy Compatibility

Institutional compatibility concerns the readiness of research organisations, funding bodies, and national policy environments to support AI-enabled field research in a sustained and equitable manner. The World

Bank's Digital Progress and Trends Report identifies four preconditions for countries to meaningfully benefit from AI: reliable and affordable connectivity, locally relevant data, a digitally skilled population, and adequate computing capacity — conditions that are unevenly distributed across the Global North and Global South and that directly shape which research communities can adopt AI-enabled field methods.

6. Challenges and Limitations

6.1 Infrastructure and the AI Divide

A recurring theme across the reviewed literature is the emergence of an 'AI divide' layered atop the pre-existing digital divide. Analyses of AI adoption in developing countries consistently identify inadequate internet and power infrastructure, limited computing capacity, and shortages of AI-skilled personnel as primary constraints, with the risk that AI-enabled field research becomes concentrated in well-resourced institutions and geographies, potentially reinforcing rather than reducing global research inequities.

6.2 Data Quality, Representativeness, and Algorithmic Bias

AI models are only as reliable as the data on which they are trained. Field settings frequently generate imbalanced, sparse, or non-representative datasets — for example, camera-trap or satellite datasets skewed toward well-studied regions or species — which can propagate bias into downstream models and, by extension, into the scientific and policy conclusions drawn from them.

6.3 Ethical, Privacy, and Governance Concerns

Field research involving human participants, indigenous communities, or protected species raises distinct governance questions regarding data ownership, consent, and the potential misuse of AI-derived location or behavioural data. Survey evidence indicates that practitioners themselves regard the absence of clear AI governance frameworks as one of the most significant barriers to trustworthy adoption in sustainability-oriented research.

6.4 The Environmental Footprint of AI Systems

A further tension, increasingly noted in the sustainability literature, is that AI itself carries a non-trivial ecological footprint, associated with the energy and water demands of model training and inference. Systematic reviews of AI for sustainability caution that its own environmental cost, along with the risk of algorithmic bias, must be proactively managed rather than treated as incidental to its benefits, since these costs can partially offset the sustainability gains that AI-enabled field research is intended to produce.

7. Discussion: Toward a Conditional Compatibility Framework

Synthesising the evidence above, this review proposes that the compatibility of AI with field research, viewed from a sustainable development perspective, is best understood as conditional rather than absolute. AI demonstrably enhances the scale, precision, and cost-efficiency of field data acquisition across environmental, agricultural, and social domains, and a substantial body of expert and bibliometric evidence links these gains to concrete SDG targets, particularly SDG 2, 3, 6, 11, 13, 14, and 15. At the same time, the same body of literature consistently identifies four countervailing forces — infrastructural

inequity, data and algorithmic bias, governance and ethical gaps, and AI's own resource footprint — that can erode these gains if left unmanaged. Compatibility, in this sense, is not a fixed property of AI technology but an outcome jointly produced by technical design choices, institutional investment, and governance frameworks. This suggests that assessments of AI's role in field research should move beyond binary judgements of adoption versus non-adoption, and instead evaluate specific AI-enabled methods against the full set of technical, economic, ethical-social, and institutional conditions outlined in Section 5.

8. Recommendations for Research and Practice

- Prioritise low-connectivity and low-power AI architectures (e.g., edge computing, on-device inference) to extend compatibility to infrastructure-constrained field sites.
- Invest in locally representative training data and participatory data-collection protocols to reduce bias and improve model validity across diverse field contexts.
- Embed informed-consent and data-governance protocols specific to AI-enabled field methods, particularly where human subjects, indigenous knowledge, or protected species are involved.
- Adopt lifecycle assessment of the energy and computing footprint of AI tools used in field research, favouring efficient models where field-level accuracy gains are marginal.
- Strengthen international research-capacity partnerships and funding mechanisms to reduce the AI divide between well-resourced and resource-constrained research institutions.
- Develop discipline-specific reporting standards for AI-assisted field studies to improve reproducibility, transparency, and comparability across the literature.

9. Conclusion

This review has examined the compatibility of artificial intelligence with field research through the lens of sustainable development, synthesising evidence from environmental monitoring, agriculture, remote sensing, and social and public-health fieldwork. The evidence indicates that AI can meaningfully advance field-research practice and contribute to multiple SDGs by improving the scale, speed, and precision of data collection and analysis. However, this compatibility is not unconditional: it depends on addressing persistent infrastructural inequities, data and algorithmic bias, governance gaps, and the environmental costs of AI systems themselves. Future research should prioritise empirical, context-specific evaluations of AI-enabled field methods — particularly in under-resourced regions — to build a more granular, evidence-based understanding of when, where, and how AI genuinely advances sustainable development goals, as opposed to merely extending existing patterns of research inequity under a new technological guise.

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